

KAIYAN ZHENG

(She/Her)

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Education

Northwestern University

Industrial Engineering and Management Sciences

Evanston, Illinois

Expected Graduation: May 2031

- **Program:** PhD in Industrial Engineering and Management Sciences

University of Michigan, Ann Arbor

Department of Mathematics, Statistics & EECS

Ann Arbor, Michigan

May 2026

- *Transferred from The Chinese University of Hong Kong, Shenzhen (2022 - 2024, Quantitative Finance)*
- **Program:** BS Mathematics - Operations Research and Modeling; BS Honors Data Science; BS Statistics
- **GPA:** 3.8/4 (James B. Angell Scholar), Major GPA: 3.9
- **Core Coursework:** Mathematical Analysis, Regression Analysis, Combinatorics, Differential Equations, Data Structures and Algorithms, Database Systems, Statistical Machine Learning
- **Selected Graduate Level Courses:** Real Analysis, Nonlinear Programming, Statistical Theory, Experimental Design, Optimization, Stochastic Processes (I & II), Stochastic Optimization and Machine Learning

Research

Dynamics of Gradient Descent via Function Basis Decomposition

Independent Research Project

May 2025 – Present

University of Michigan

- Under supervision of *Prof. Salar Fattahi* at the Department of Industrial and Operations Engineering.
- Characterize the behavior gradient descent via basis function decomposition under multiple statistical settings. Extending gradient-descent convergence analysis from the population to the stochastic and empirical case.

Dynamic Evacuation Recommendation with Inferred Trust

Independent Research Project

Jan 2025 – Present

University of Michigan

- Under supervision of *Prof. Ruiwei Jiang* at the Department of Industrial and Operations Engineering.
- Developed a real-time evacuation route recommendation system, solving a dynamic optimization problem under trust modeling and inference. Demonstrated that the trust-inferred recommendation system outperforms the non-recommendation baseline in key metrics.
- Submitted to 2026 IEEE Conference of Decision and Control

Federated Unlearning with Orthogonal Steepest Descent

Research Project with Zibin Pan et. al.

May 2024 – Aug 2024

CUHK(SZ)

- At the Laboratory of Energy Internet under *Prof. Junhua Zhao*, focusing on privacy-preserving machine learning and optimization in federated systems.
- Co-designed and implemented the FedOSD federated unlearning algorithm, introducing a stabilized unlearning loss and confirming minimal overhead of the SVD-based orthogonalization step. Implemented the full empirical evaluation under diverse client distributions and contributed to the AAAI 2025 manuscript.

Publication

Federated Unlearning with Gradient Descent and Conflict Mitigation

Zibin Pan, Zhichao Wang, Chi Li, Kaiyan Zheng, Boqi Wang, Xiaoying Tang, Junhua Zhao

AAAI 2025

Paper on AAAI

Teaching Service

Grader

Feb 2026 – May 2026

IOE 511: Continuous Optimization Methods

University of Michigan

- Supervisor: Prof. Albert Berahas at Department of Industrial and Operations Engineering

Academic Mentor and Coordinator

Feb 2024

Freshmen Winter Camp of CUHK(SZ)

School of Data Science, CUHK(SZ)

- Conducted lectures to 250+ students on introductory machine learning as an Academic Mentor. Planned and managed the lecture schedule and materials as an Academic Coordinator.

Undergraduate Student Teaching Fellow

Sep 2023 – Dec 2023

CSC 1001: Introduction to Computer Science - Python Programming

School of Data Science, CUHK(SZ)

- Engaged in homework grading design, led weekly lab sessions, held office hours, for a Python programming course with 1000+ students.
- Supervisor: Prof. Junhua Zhao at School of Science and Engineering

Awards

- University of Michigan, James B. Angell Scholar 2026
- University of Michigan, University Honors 2024–2026
- CUHK(SZ) Bowen Scholarship, Dean's List 2022–2024
- Second Prize, China Undergraduate Mathematical Contest in Modeling 2023
- First Prize, China National High School Biology Olympiad 2021

Presentations

- Summer Undergraduate Research in Engineering (SURE) Program Showcase Jul 31, 2025
- UM-Mathematics Directed Reading Program Showcase Aug 28, 2025
- UM-IOE Undergraduate Research Symposium Sep 19, 2025
- UM-IOE 70th Anniversary Symposium Oct 3, 2025
- UM Statistics Undergraduate Research Symposium Apr 17, 2026

Course Projects

Population Trend Analysis and Projection Using Machine Learning

UMich

DATASCI 306 - Statistical Computing

Project on Github

- Selected as one of 3 excellent projects out of 50 groups. Led a team of 5 for a comprehensive statistical analysis and forecasting project using R, addressing global population trends.
- Implemented a Prophet machine learning model with external regressors to forecast population dynamics through 2100 with interactive visualizations.

Course Registration & Recommendation System Powered by LLM and TTS

CUHK(SZ)

CSC 3170 - Database Systems

Project on Github

- Coordinated a cross-functional team to design and implement an advanced SIS, addressing inefficiencies in academic management systems.
- Contributed to the integration of pre-trained Large Language Models (LLMs) for personalized course recommendations and implemented Zero-Shot Text-to-Speech (TTS) models to enhance accessibility.

Skills

- **Programming Languages:** Python · C&C++ · Java · MATLAB · R · SQL · $\LaTeX$
- **Test Scores:** GRE: 328 (Quantitative 170; Verbal 158; Analytical Writing 3.5)
- **Frameworks:** PyTorch · Tensorflow · NumPy · pandas · scikit-learn · CVX · Gurobi · streamlit
- **Hobbies:** I love art, painting and drawing! (Artwork Portlio)